

Temporal genic context of acetylation peaks (0–24 hr)

We annotated peak coordinates against mm10 gene models to quantify their distribution across promoters (TSS-proximal regions), exons, introns, 5'/3' UTRs, and an “other” category at 0, 6, and 24 hr.

At 0 hr (Figure 1), all three marks are dominated by intronic assignments; exonic overlap is modest and 5'/3' UTRs are rare. Promoter-associated peaks are comparatively more abundant for H3K27ac and H4K5ac than for H2BK12ac, which shows a reduced promoter fraction and an elevated “other” component.

At 6 hr (Figure 2), the intron-dominant pattern persists across marks. H3K27ac and H4K5ac retain similar promoter fractions, whereas H2BK12ac continues to show a smaller promoter share with a larger “other” category. Exonic contributions remain modest and UTRs minimal.

By 24 hr (Figure 3), introns remain the principal feature class for all marks. Promoter representation is highest for H4K5ac, intermediate for H3K27ac, and lowest for H2BK12ac; exonic overlap remains modest, UTRs minimal, and “other” assignments substantial across marks.

Collectively, acetylation signal localizes predominantly within introns over 0–24 hr, with H2BK12ac consistently depleted at promoters relative to H3K27ac and H4K5ac at every time point.

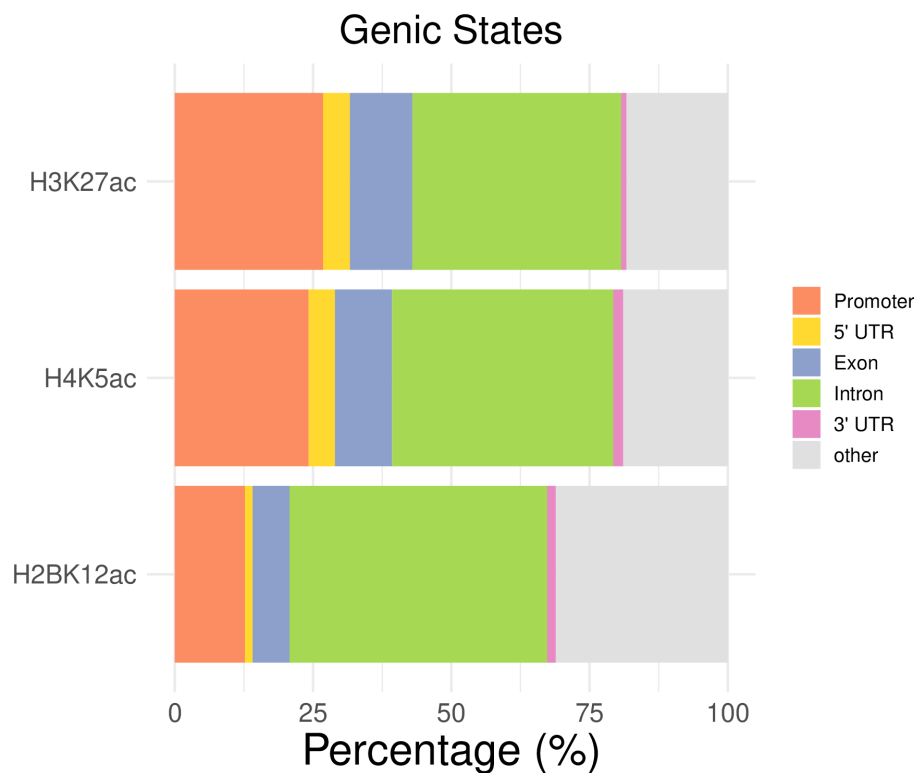


Figure 1. Genomic annotation distribution of 0hr histone mark peaks. Peaks for H3K27ac, H4K5ac, and H2BK12ac were assigned to promoters, UTRs, exons, introns, or other regions at 0hr. Introns and other regions dominate across marks; promoter-associated peaks are relatively higher for H3K27ac and H4K5ac than for H2BK12ac, while UTR contributions are minimal.

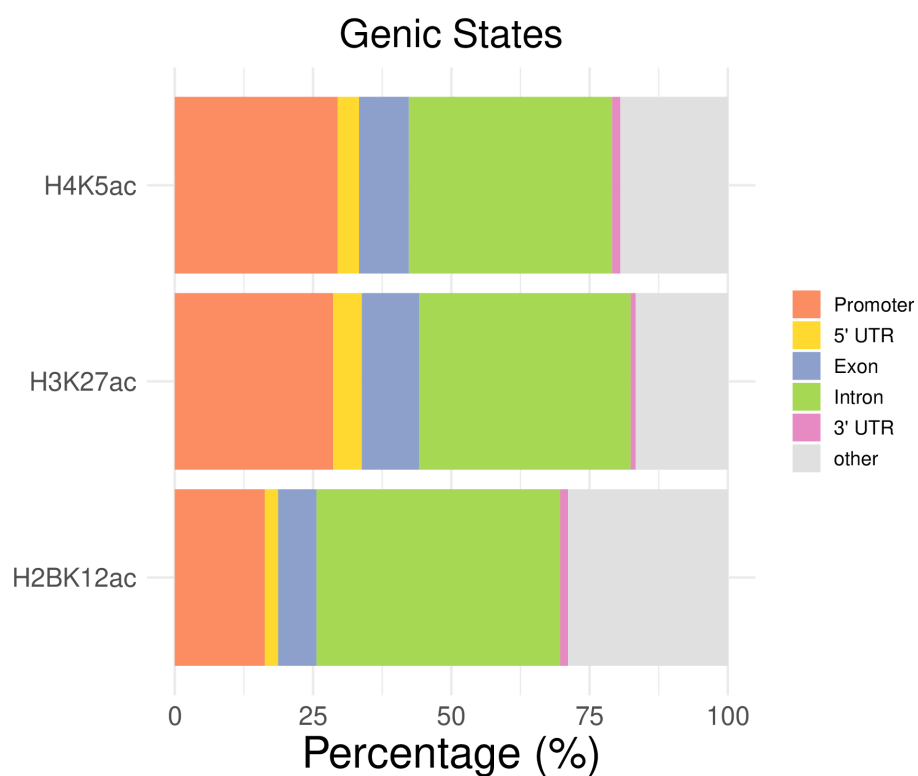


Figure 2. 6hr peak distribution across genic features for three acetylation marks. Peaks from *H4K5ac*, *H3K27ac*, and *H2BK12ac* at 6hr were annotated to promoters, UTRs, exons, introns, and other categories. Signals are primarily intronic with secondary promoter contributions; *H2BK12ac* exhibits less promoter assignment and relatively more in other regions compared with *H4K5ac* and *H3K27ac*.

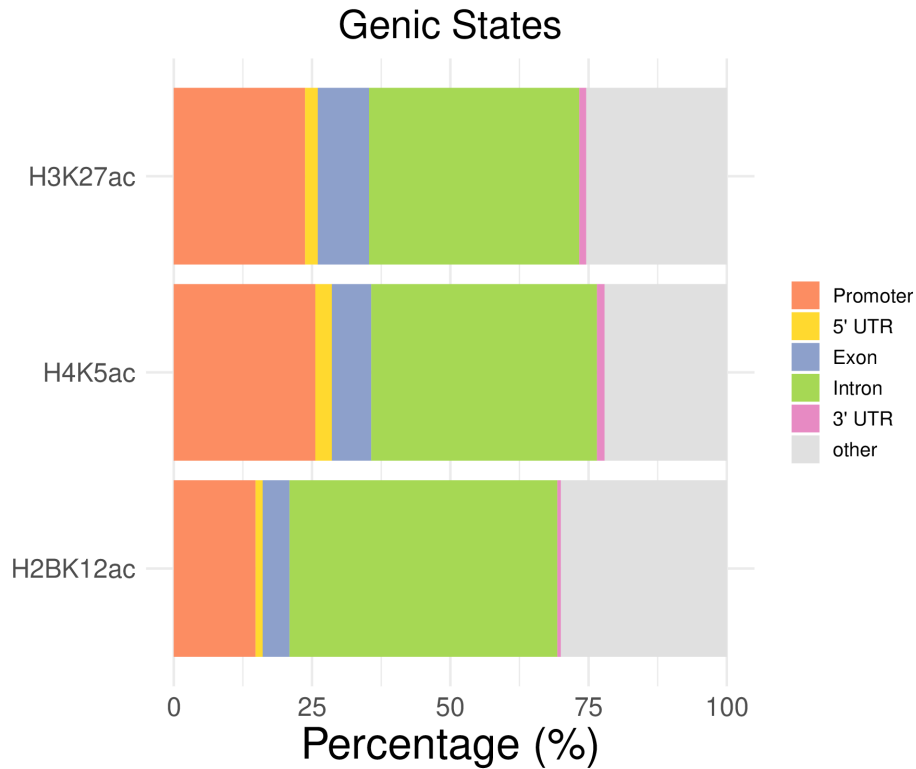


Figure 3. Genic distribution of 24hr peaks across histone acetylation marks. Percentages of 24hr peaks overlapping gene features are shown for H3K27ac, H4K5ac, and H2BK12ac. Most peaks fall within introns with a secondary promoter fraction, while exons and UTRs are minor; H2BK12ac is most intron-skewed and H4K5ac shows relatively more promoter overlap.

Regulatory element class shifts via ENCODE cCRE overlap across time

To contextualize regulatory activity over time, we intersected peak sets at 0, 6, and 24 hr with ENCODE candidate cis-regulatory element (cCRE) classes, partitioning overlaps into promoter-like (PLS), enhancer-like (proximal pELS and distal dELS), chromatin-accessibility subclasses (CA-H3K4me3, CA-CTCF, CA-TF), and non-annotated (“other”) regions (Figures 4–6).

For H3K27ac and H4K5ac, enhancer-like cCREs dominate at all time points, with dELS and pELS comprising the major fractions and PLS contributing secondarily (Figures 4–6). CA subclasses, including CA-H3K4me3, account for only small portions, and CA-CTCF/CA-TF remain minimal. The enhancer-to-promoter balance is stable from 0 to 24 hr, indicating that these acetylation marks principally track active enhancer landscapes with a persistent, smaller promoter component.

H2BK12ac shows a more dispersed allocation across cCRE classes at each time point: while dELS is retained, pELS and PLS are comparatively reduced, and CA categories and “other” regions are expanded (Figures 4–6). This pattern is evident at 0 and 6 hr and is

most pronounced at 24 hr, which exhibits the largest “other” fraction among the marks. CA–CTCF and CA–TF remain minor across all marks and times. Together, these results underscore enhancer-centric engagement for H3K27ac/H4K5ac, whereas H2BK12ac is less

anchored to canonical promoter/enhancer cCREs and more frequently occupies broader CA or unannotated loci over 0–24 hr.

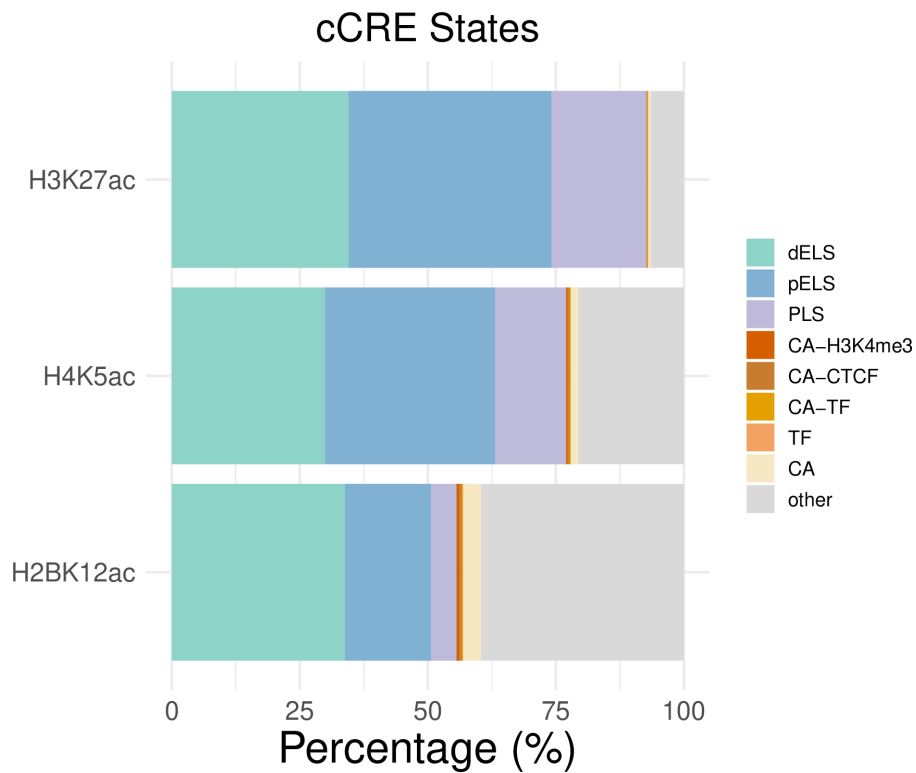


Figure 4. Peaks partitioned across ENCODE cCRE states by mark. *Overlap of called peaks with ENCODE cCRE categories is summarized for H3K27ac, H4K5ac, and H2BK12ac. Enhancer-like elements (dELS, pELS) predominate for H3K27ac and H4K5ac, whereas H2BK12ac shows increased CA and other with reduced PLS; CTCF- and TF-associated components are minor.*

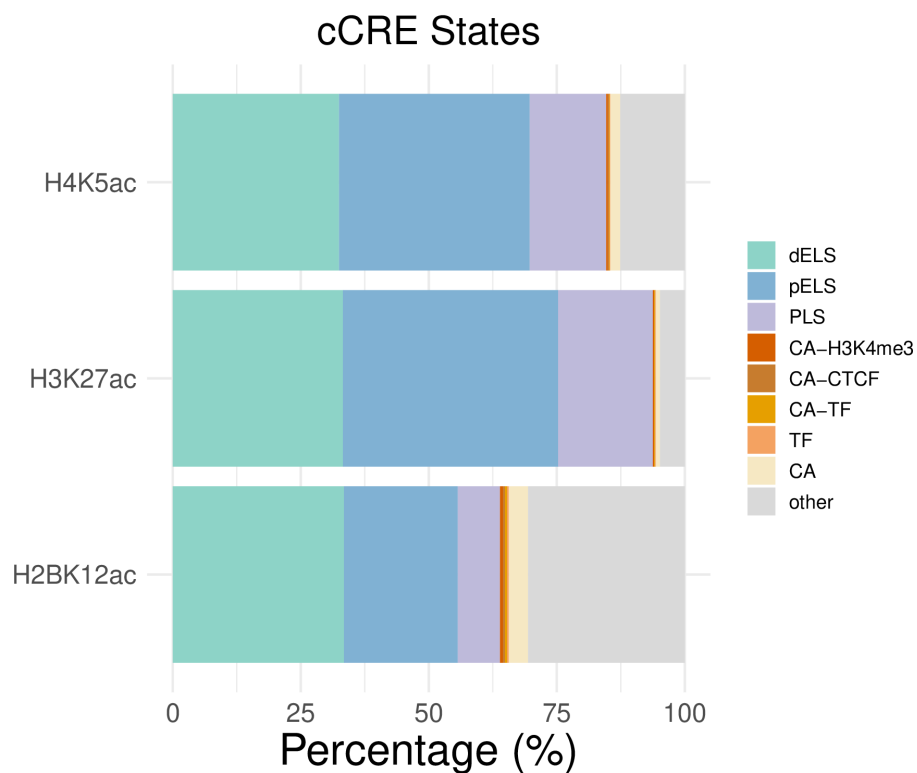


Figure 5. 6hr peaks across cCRE classes for three acetyl marks. Percent distributions of cCRE categories overlapped by 6hr peaks are shown for H4K5ac, H3K27ac, and H2BK12ac. pELS and dELS dominate for H4K5ac and H3K27ac, whereas H2BK12ac has a larger CA/other share and smaller promoter-like segments. CTCF- and TF-associated elements are minor in all three.

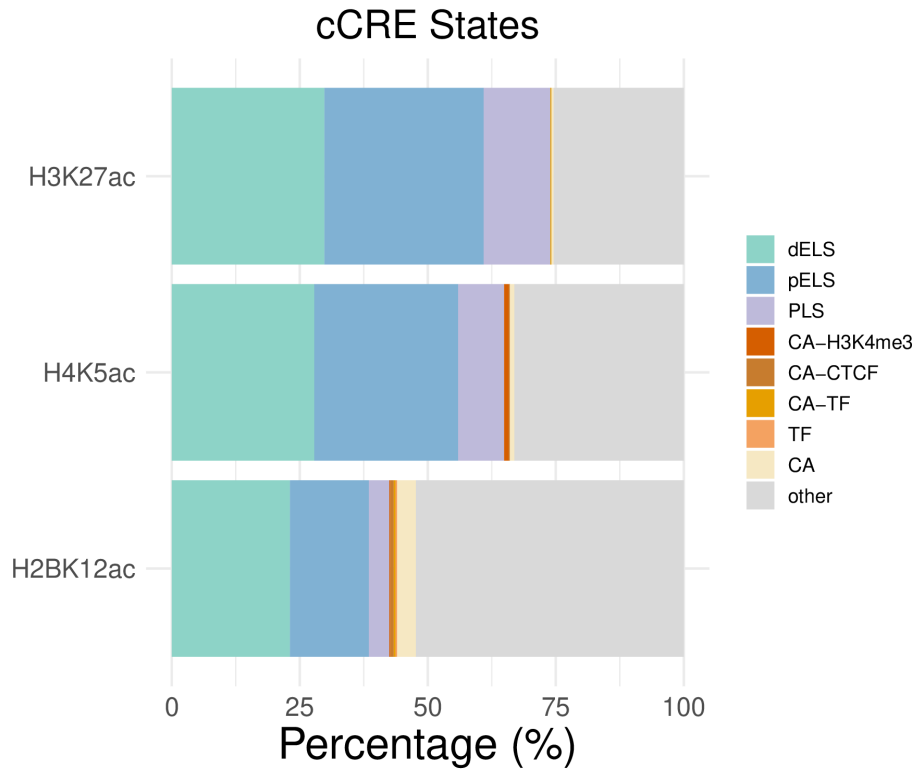


Figure 6. 24hr peak distribution across ENCODE cCRE categories. Proportions of peak overlaps are shown for H3K27ac, H4K5ac, and H2BK12ac. Enhancer-like states (dELS, pELS) dominate for H3K27ac and H4K5ac, with smaller PLS and minimal CTCF/TF-associated elements. H2BK12ac is characterized by a larger “other” share and reduced enhancer-like overlap.

Transcription factor motif activity dynamics from baseline to 24 hr

We profiled transcription factor (TF) motif enrichment across acetylation-defined peak sets at 0 hr (baseline), 6 hr, and 24 hr to nominate candidate regulators and assess mark specificity (Figures 7–9). Three consistent features emerged: (i) motif loads are markedly stronger in H3K27ac (and generally H4K5ac) than in H2BK12ac at all time points; (ii) families centered on EP300, ETS/IRF, AP-1/MYC, ELF1, and TBP are recurrently over-represented; and (iii) H3K27ac shows the clearest temporal amplification of inducible TF modules.

At 0 hr (Figure 7), H3K27ac and H4K5ac already harbor broad enrichment for EP300; AP-1/MYC components (JUND, JUN, MAX, MXI1/MYC); ELF1; and TBP, with additional signals for UBTF, USF1/2, and POLR2A/phospho-associated motifs. Architectural factors—CTCF and cohesin components (SMC3, RAD21)—are moderate in H2BK12ac and stronger in H3K27ac/H4K5ac. Several motifs (e.g., PAX5, E2F4, POU5F1) are weak or negative in H2BK12ac yet remain positive in the histone acetylation marks, highlighting the greater regulatory specificity of H3K27ac/H4K5ac at baseline.

By 6 hr (Figure 8), enrichment intensifies most prominently in H3K27ac, with EP300, ETS1, IRF4, ELF1, MAX/MXI1/MYC, MEF2A, TBP, and JUN/JUND showing pronounced gains. H2BK12ac remains comparatively low across most TFs, whereas factors such as MAFK and GABPA maintain moderate, more mark-agnostic enrichment.

At 24 hr (Figure 9), H3K27ac and, to a lesser extent, H4K5ac sustain or further increase enrichment for ETS1, IRF4, ELF1, and AP-1 family members, with GABPA showing a particularly strong H3K27ac signal. Transcriptional machinery-associated motifs (e.g., POLR2AphosphoS5) and pluripotency-associated motifs (e.g., POU5F1) remain weak or near baseline in H2BK12ac yet are elevated in H3K27ac. CTCF/cohesin-related motifs (CTCF, RAD21) are notable in H3K27ac/H4K5ac at 0 and 24 hr.

Integrated over time and marks, these patterns nominate a compact regulatory axis. EP300—both as a highly enriched motif and as the principal H3K27 acetyltransferase—implicates enhancer-centric activation, consistent with its role in depositing H3K27ac at active regulatory regions (Zhang et al., 2026). Coordinated enrichment of ETS and IRF motifs (ETS1, ELF1, IRF4) indicates an ETS–IRF composite module characteristic of stimulus-evoked programs (Katko et al., 2026). Parallel over-representation of AP-1/MYC circuitry (JUN/JUND/MAX/MYC/MXI1) points to immediate-early and growth-associated control that persists through 24 hr, and TBP enrichment reflects sustained core promoter engagement accompanying inducible enhancer activity. H3K27ac consistently provides the highest-amplitude readout of these inducible modules (with H4K5ac tracking at lower magnitude), whereas H2BK12ac shows weaker, more variable motif enrichment, suggesting broader chromatin acetylation with less TF-tethered enhancer selectivity. The repeated detection of CTCF/cohesin motifs within H3K27ac/H4K5ac contexts at 0 and 24 hr is compatible with acetylation-associated activity at regulatory boundaries; more generally, histone hyperacetylation can differentially modulate intra-TAD interactions versus CTCF-anchored loops (Smith et al., 2026). Together, these data highlight EP300, ETS/IRF (ETS1/ELF1/IRF4), AP-1/MYC (JUN/JUND/MAX/MYC), TBP, and late-

appearing GABPA as core candidates, preferentially loaded in H3K27ac (and secondarily H4K5ac) peaks.

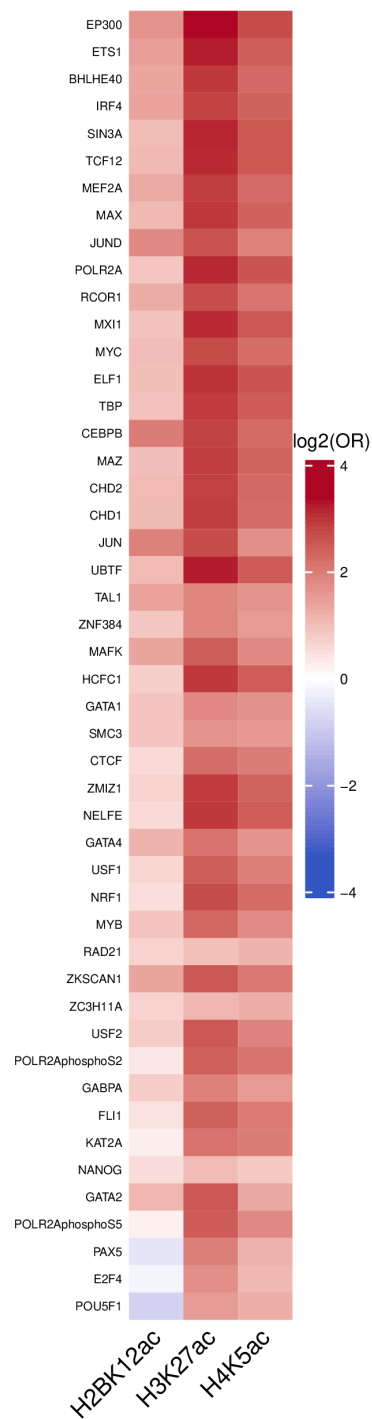


Figure 7. TF motif enrichment across acetylation-marked peaks at 0 hr. Motif over-representation is summarized for peaks associated with H2BK12ac, H3K27ac, and H4K5ac in the 0 hr sample. H3K27ac and H4K5ac show broadly stronger enrichment than H2BK12ac; examples with high signals include EP300, UBTF, and TCF12, whereas E2F4 and POU5F1 are weak or depleted mainly in H2BK12ac.

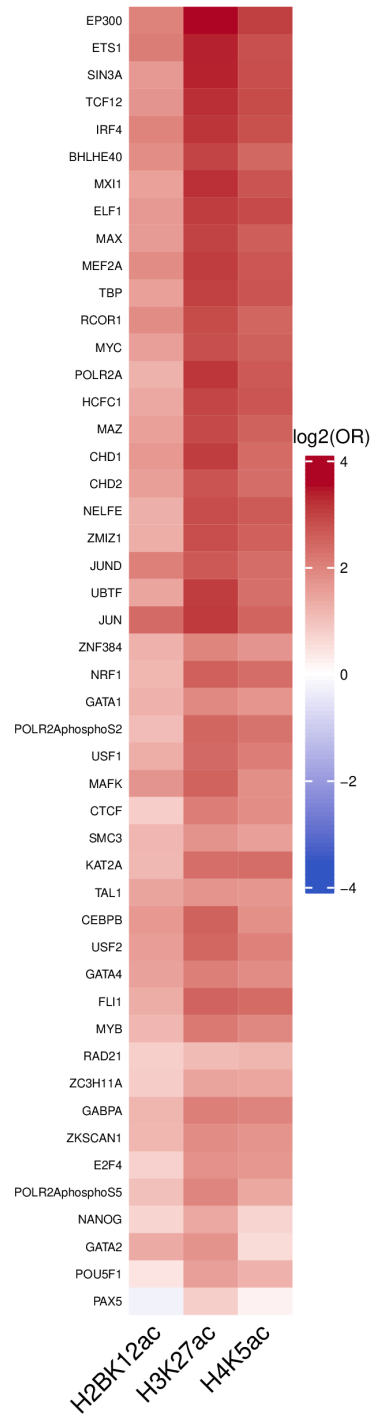


Figure 8. Motif enrichment across 6hr histone acetylation peaks. Heatmap of log2 odds-ratio enrichment for transcription factor motifs within peaks of H2BK12ac, H3K27ac, and H4K5ac at 6hr relative to background. H3K27ac shows the strongest overall enrichment across factors, whereas H2BK12ac and H4K5ac are more moderate with several motifs near baseline.

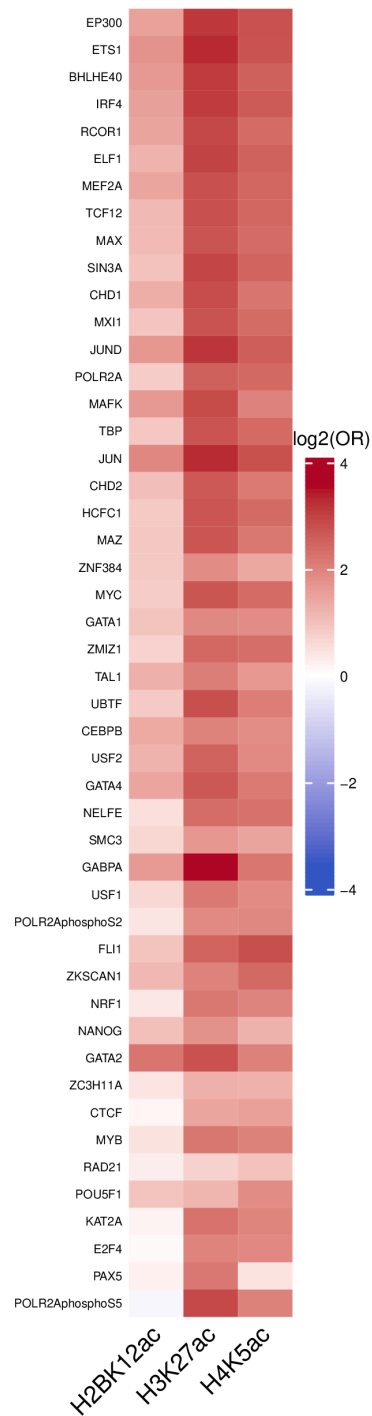


Figure 9. TF motif enrichment across 24hr acetylation-mark peaks. Motif enrichment (log2 odds ratio) is summarized for peaks called in the 24hr sample across H2BK12ac, H3K27ac, and H4K5ac. Enrichment is broadly higher for H3K27ac and H4K5ac, with a particularly strong signal for GABPA and consistently elevated motifs such as JUN, IRF4, and ETS1, while several motifs are weaker in H2BK12ac.

Pathway enrichment dynamics across marks and time points (0–24 hr)

Across 0–24 hr, pathway enrichments reveal a transition from a broad H2BK12ac-dominated baseline to focused, enhancer-linked immune activation driven by H3K27ac, culminating in selective maintenance of IFNG/TNFA axes by 24 hr. At 0 hr (Figure 10), enrichment is strongly polarized to H2BK12ac and spans diverse stress, proliferative, and immune programs, including APOPTOSIS, COMPLEMENT, ESTROGEN_RESPONSE (early/late), HYPOXIA, IL2_STAT5_SIGNALING, IL6_JAK_STAT3_SIGNALING, INTERFERON_GAMMA_RESPONSE, INFLAMMATORY_RESPONSE, TNFA_SIGNALING_VIA_NFKB, MTORC1_SIGNALING, P53_PATHWAY, MITOTIC_SPINDLE, UV_RESPONSE_UP, and XENOBIOTIC_METABOLISM, with minimal signal from H3K27ac or H4K5ac. This baseline suggests a permissive H2BK12ac-biased chromatin state encompassing both immune and non-immune hallmarks.

By 6 hr (Figure 11), the enrichment landscape pivots to H3K27ac, with the strongest signals concentrated in immune/cytokine pathways—IL2_STAT5_SIGNALING, IL6_JAK_STAT3_SIGNALING, INFLAMMATORY_RESPONSE, INTERFERON_ALPHA_RESPONSE, INTERFERON_GAMMA_RESPONSE, and TNFA_SIGNALING_VIA_NFKB—while MTORC1_SIGNALING and P53_PATHWAY also remain prominent. Metabolic and cell-cycle programs are largely absent across marks. In contrast, H2BK12ac shows only isolated enrichment (e.g., ADIPOGENESIS) and H4K5ac remains uniformly weak. The emergence of H3K27ac-centric immune/cytokine signatures is consistent with its canonical role in activating inflammatory enhancers (Lu et al., 2025) and with the broader contribution of histone acetylation to immune gene regulation (Quintero-Ronderos & Montoya-Ortiz, 2012).

At 24 hr (Figure 12), enrichment consolidates into a narrow immune set. IL6_JAK_STAT3_SIGNALING remains strongest with H3K27ac, whereas INTERFERON_GAMMA_RESPONSE and TNFA_SIGNALING_VIA_NFKB are most prominent with H4K5ac. Most other pathways, including INTERFERON_ALPHA_RESPONSE, APOPTOSIS, and MYC_TARGETS, are uniformly weak across marks, and H2BK12ac shows no notable enrichment. Thus, an early, broad H2BK12ac baseline resolves into an H3K27ac-driven immune/cytokine program by 6 hr and a selectively sustained IFNG/TNFA axis by 24 hr, with H2BK12ac signals generally waning after baseline. These dynamics align with enhancer-mediated

activation of inflammatory circuits marked by H3K27ac and the temporal sharpening of cytokine signaling outputs (Lu et al., 2025; Quintero-Ronderos & Montoya-Ortiz, 2012).

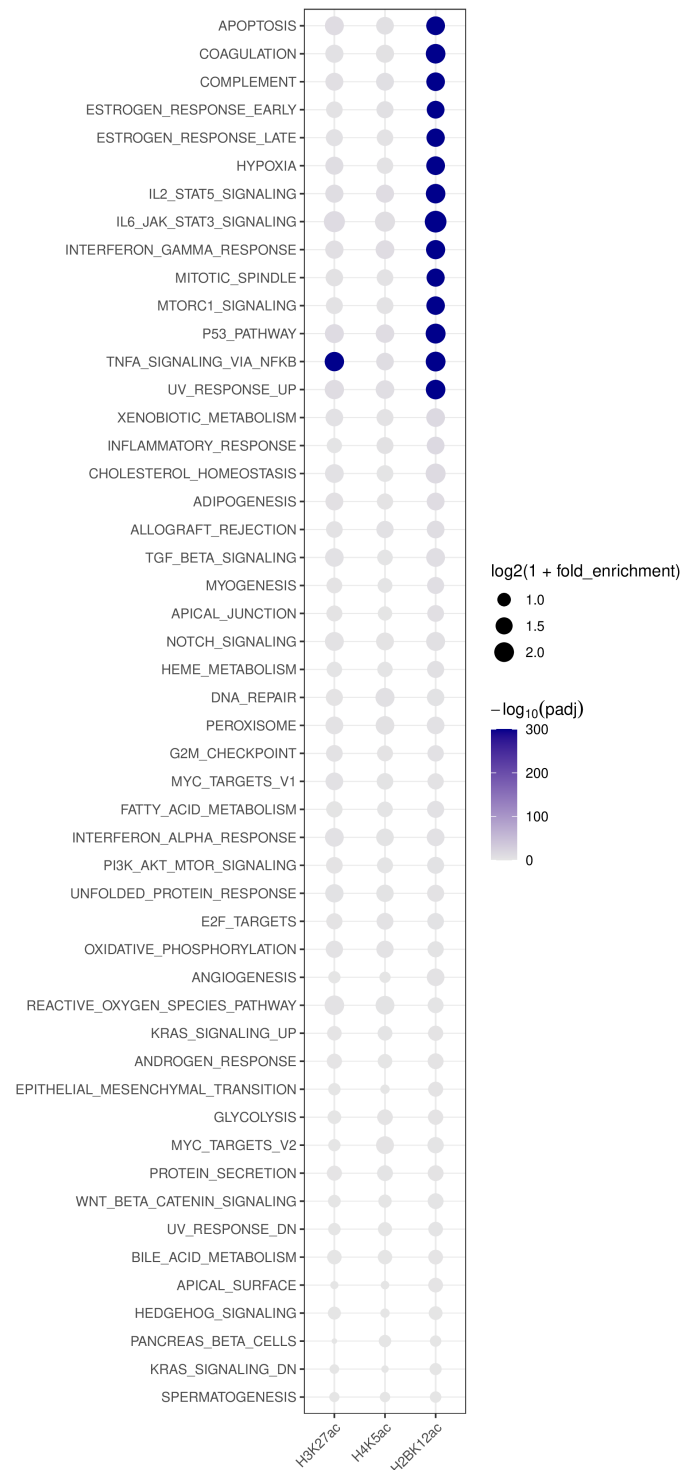


Figure 10. Pathway enrichment at 0hr is concentrated in H2BK12ac. Enrichment analysis of genes linked to peaks in the 0hr sample is shown for three histone marks (H3K27ac, H4K5ac, H2BK12ac). Stronger associations are largely restricted to H2BK12ac, spanning stress, immune, and proliferation pathways such as APOPTOSIS, INTERFERON_GAMMA_RESPONSE, TNFA_SIGNALING_VIA_NFKB, P53_PATHWAY, ESTROGEN_RESPONSE_EARLY/LATE, and HYPOXIA, while H3K27ac and H4K5ac show minimal enrichment.

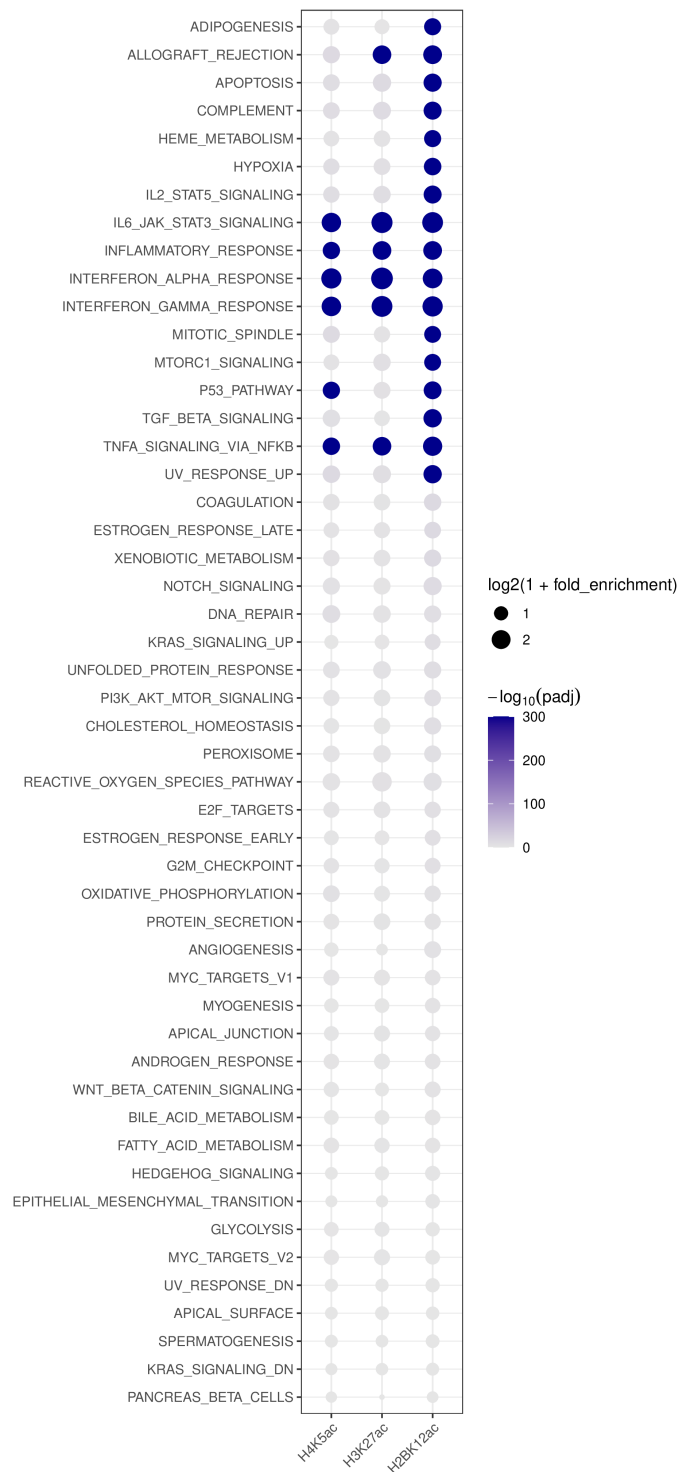


Figure 11. 6hr pathway enrichment emphasizes immune and cytokine signaling. *Enrichment of Hallmark pathways was computed for genes linked to 6hr peaks across three histone acetylation marks. Signals concentrate in H3K27ac, dominated by immune and inflammatory programs (IL2/STAT5, IL6/JAK/STAT3, interferon, TNFA/NFKB), whereas H4K5ac is broadly weak and H2BK12ac shows limited, isolated categories.*

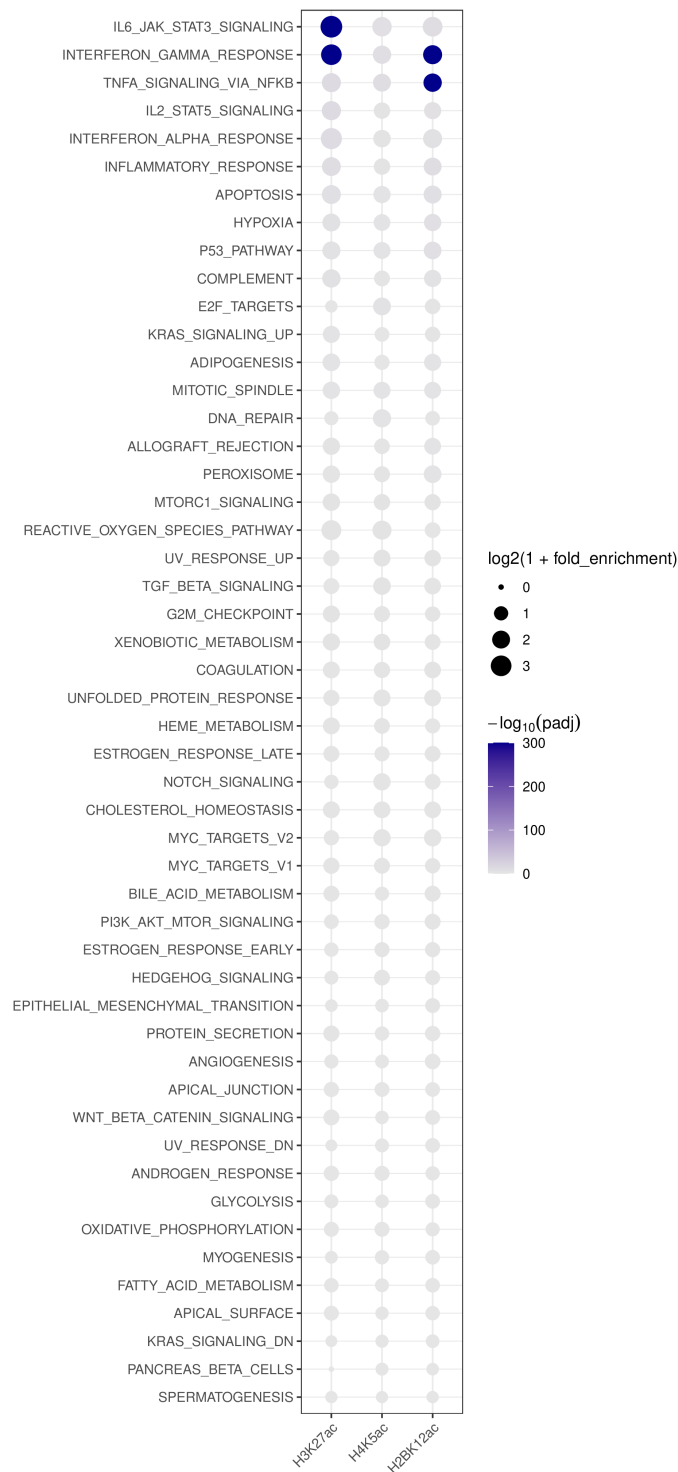


Figure 12. Selective immune pathway enrichments at 24hr across histone marks. Pathway enrichment for genes linked to 24hr peaks is shown across H3K27ac, H4K5ac, and H2BK12ac. Enrichment is concentrated in IL6_JAK_STAT3_SIGNALING for H3K27ac and in INTERFERON_GAMMA_RESPONSE and TNFA_SIGNALING_VIA_NFKB for H4K5ac, with minimal signals elsewhere and no prominent enrichments for H2BK12ac.

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